

Alexander J. Sampson

Department of Mathematics | Florida State University

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EDUCATION

Ph.D. in Applied and Computational Mathematics

Florida State University – Tallahassee, FL

Advisor (informal): Prof. Martin Bauer 2025 – present

Graduate Study in Applied Mathematics

University of Nevada, Las Vegas – Las Vegas, NV

2021 – 2024

B.A. in Applied Mathematics, minor in Physics

University of Montana, Missoula, MT

2014 – 2021

RESEARCH INTERESTS

My interests span the boundary between pure and applied mathematics, with a focus on the analytic and geometric foundations of applied theory — particularly shape analysis, PDE theory, and optimal transport, with applications ranging from data science and biology to mathematical physics. Current areas of focus include:

- **Shape analysis & infinite-dimensional geometry:** Riemannian geometry of shape spaces; geometric PDEs on spaces of curves and surfaces.
- **Optimal transport & Wasserstein geometry:** Analytical and geometric frameworks for shapes, signals, and distributions.
- **PDEs in applications:** Qualitative properties of solutions to models from physics and biology.
- **Variational & topological methods:** Weak solution frameworks, energy methods, and topological degree theory.

CURRENT RESEARCH

I am a first-year Ph.D. student at FSU, currently reading toward a project in *shape analysis and infinite-dimensional Riemannian geometry* under the informal supervision of Prof. Martin Bauer. My reading focuses on:

- M. Bauer, C. Maor, B. Wirth, “Completeness of reparametrization-invariant Sobolev metrics on the space of surfaces,” *arXiv:2512.01566* (2025).

TEACHING EXPERIENCE

Instructor of Record — University of Nevada, Las Vegas

Pre-Calculus II Summer 2023, 2024

Graduate Assistant & Discussion Instructor — UNLV

2021 – 2024

• Online course development | Office of Online Education | Pre-Calculus II & Finite Mathematics:

Produced lecture notes and study guides, developed instructor manuals for course transitions, and responded to student questions on course content.

- **Discussion instructor** | Calculus I, II, & III; Discrete Mathematics; ODEs for Engineers:

Developed and graded weekly quizzes and homework, and independently taught weekly sessions reviewing material from the main lecture.

Private Tutor — Independent

2015 – present

Subjects: High School Mathematics, College Algebra, Trigonometry, Pre-Calculus, Calculus I/II, Linear Algebra, Vector Calculus.

SELECTED GRADUATE COURSEWORK

University of Nevada, Las Vegas

Numerical Methods I & II
Measure Theory
Partial Differential Equations
Advanced ODEs I & II
Mathematical Physics I
Abstract Algebra III

Real Analysis I
Complex Function Theory II
Advanced PDEs I & II
Spectral Theory
Nonlinear Analysis & Degree Theory
Methods in Numerical Analysis

Florida State University

Methods of Applied Mathematics I & II
Foundations of Computational Mathematics I & II

Elementary PDEs I & II

UNDERGRADUATE BACKGROUND

- **Applied Mathematics** (University of Montana): Calculus I–III, Linear Algebra, Abstract Algebra, ODEs, PDEs, Dynamical Systems, Advanced Mathematical Methods I & II, Analysis for Applied Mathematicians, Deterministic Models, Introduction to Mathematical Software.

- **Physics** (minor): Fundamentals of Physics I & II, Modern Physics, Intro to Theoretical Physics, Classical Mechanics, Electrodynamics I & II, Quantum Mechanics I, Thermodynamics & Statistical Mechanics, Intro to Plasma Physics.

CONFERENCES

- Joint Alabama-Florida (JAF) Conference and Workshop on Differential Equations, Dynamical Systems and Applications
Florida International University (FIU) | Miami, FL • Attendee, travel support received May 2026
- AMS Spring Southeastern Sectional Meeting
Georgia Southern University–Armstrong Campus | Savannah, GA • Attendee March 2026
- Joint Alabama–Florida Conference on Differential Equations, Dynamical Systems and Applications (JAF DEDS)
Auburn University | Auburn, AL • Attendee, travel support received May 2023

SKILLS

Computing MATLAB, $\text{\LaTeX}$

SERVICE & LEADERSHIP

Vice President, UNLV SIAM Student Chapter 2023 – 2024

Scheduling Liaison / Secretary, UNLV SIAM Student Chapter 2022 – 2023